DIVYANSHU GOYAL

Multimodal Model Training | RL Fine-Tuning | Evaluation
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SUMMARY

Applied scientist at Adobe, passionate about multimodal large language models. Over the past 5 years I have trained, aligned, and evaluated these models in highly subjective domains such as brand marketing and photo editing/retouching. I have contributed my work at CVPR, ICML, and AAAI workshops, hold patents in generative AI space, and write about AI through blogs. I am strong believer in first principle thinking and try to bring that to everything I do.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

VLM Fine-Tuning with SFT and Reinforcement Learning (Gemma-3-12B)

- Built distributed **SFT** and multi-round **GRPO** pipelines to train Gemma-3-12B (12.2B parameters; LoRA rank 256, 1.19B trainable) to assess marketing-image quality on 8× H100: each RL round samples a group of completions per prompt, scores them with programmatic rewards against ground truth, and trains with a clipped policy-gradient loss.
- Audited GRPO runs for **reward hacking** and found the policy learning safe guesses: over 20 rounds, one magnitude word grew from 55% to a peak of 89% of its answers, and in another run it gave the middle quality rating to 84% of images when reviewers gave it to 56%. Traced the causes to the reward's shape, an unlearnable label, and lost exploration, and redesigned the scorers.

Multimodal Judge Models

- Built a 1.1M-row training set from 14 human-rated sources with rater disagreement as soft labels, and trained **Qwen3.5-4B** LoRA judges on mixes sampled from it. With a score readout refit on crowd labels, the served model raised validation SRCC over the reference model from **0.421 to 0.663** on KonIQ and from 0.395 to 0.595 on KADID (paired 95% CIs above 0).
- Built a held-out **benchmark suite**: 8 sets, 67,168 items, leak-checked with SSCD against all 537,282 training images, with a pre-registered bootstrap non-inferiority release gate.

Unsupervised Photo Retouching (ATune)

- Led **ATune** (co-first author, under review): a 54M-parameter model that learns photo retouching from 10M unedited photographs with no expert edits and operates off-the-shelf editors by setting their sliders. With 157 raters, its edits beat GPT-6 Astra using the same editor (**54.3%** of decisive votes, 95% CI 50.3–58.3; 3,524 answers) and tied with Claude Opus 5 (50.5%, CI 46.8–54.2).
- Evaluated ATune on FiveK, PPR10K, and SPAQ with 9 published judges and Holm-corrected paired tests: on every set, at least **8 of 9** prefer it to GPT-6 Astra and at least 7 of 9 to seven learned retouchers.

Evaluation Research

- Created **DistortBench** (first author; CVPR 2026 and ICML 2026 Workshops): 13.5K questions over 27 distortion types and 5 severity levels. Of 18 VLMs, the best (Qwen3.5-27B) scores **61.9%** (chance 25%), above the average annotator (60.2%) and below a 3-annotator majority vote (65.7%).
- Co-authored **JudgeBreak** (ICML 2026 Workshop): across 95 subjective settings, linear probes on Qwen3-32B hidden states with cross-validated layer selection reach **78.1%** mean pairwise accuracy vs. 70.1% for the best reward model or zero-shot GPT-5.4 judge.

Product

- Prototyped **Adobe GenStudio for Performance Marketing** (launched October 2024, serving Fortune 500 clients) and built a multimodal **Photoshop agent** that edits images using 49 tools with vision grounding.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Co-developed Adobe's production **ML prediction service** (3,500 QPS/node at 0.3–0.6 ms p99; Adobe Tech Blog) and its model lifecycle platform.

PUBLICATIONS

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

PROJECTS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch 2026 | code
Decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with one-knob depth scaling and a custom distributed **Muon** optimizer (ZeRO-2 state sharding), trained in BF16 with FlashAttention-3 at over 1M tokens/sec on 8× H100. Fit isoFLOP curves across 40+ runs (44M to 522M parameters) to find compute-optimal model sizes. Write-ups: Going Fast and A Lazy Engineer's Guide to Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS 2026 | App Store
iPhone app that groups faces and finds duplicates on-device, with a per-build regression gate: **99.4%** BCubed F1 grouping LFW identities among 1,000 distractors, 98.0% face-detection precision on WIDER FACE.

PATENTS

Guiding Language Translation with Translation Documents Using ML (granted) 2026 | US20260050753

Brand-Aligned Marketing Content Generation 2025 | US20250117829

Cultural Adaptation of Images Using Generative AI 2026 | US20260051084

Parameter-Efficient Fine-Tuned Models 2025 | US20250217193

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Machine Learning), GPA 4.0 (2019 – 2021)

BITS Pilani – B.E. Computer Science, M.Sc. Mathematics, GPA 8.75/10 (2011 – 2016)